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Executive summary

This report presents an empirical volatility forecasting methodology aimed at producing one-month-ahead equity volatility estimates using high-frequency price data. The analysis is designed to deliver interpretable volatility forecasts that can be directly incorporated into a broader risk management and portfolio construction framework.

The estimation approach is based on the Heterogeneous Autoregressive model for realised volatility (HAR-RV). Realised variance is first constructed from five-minute intra-day returns, proxying for latent volatility while mitigating microstructure noise. The HAR-RV model then exploits the persistence of volatility across daily, weekly, and monthly horizons by regressing current daily realised variance on its lagged components. Forecasts are generated using an expanding-window, walk-forward procedure to ensure that only information available at the time of each forecast is used, thereby avoiding look-ahead bias. The key assumptions underlying the analysis include the stationarity of realised variance at relevant horizons and stable volatility dynamics over the sample period. Unit root tests support the stationarity assumption for daily, weekly and monthly realised variance measures.

The final volatility estimates indicate relatively stable one-month-ahead volatility for most assets, with only STK_E exhibiting a persistent upward trend consistent with its recent realised volatility dynamics. Notable limitations include the exclusion of trading volume and overnight price jumps, which may lead to understated volatility during periods of elevated information flow. These limitations point to natural extensions, such as augmented HAR specifications or jump-adjusted models, which could further enhance forecast accuracy.

Contents

1 Introduction

This report aims to conduct an in depth empirical analysis for forecasting one-month-ahead equity volatility, using high-frequency prices from multiple assets. The resulting forecasts are designed to serve as inputs for quant researchers and portfolio managers to use within a broader risk management and investment framework.

One of the main characteristics of volatility is the fact that it is not directly observable, even after the fact. Therefore, it is a latent variable that can only be inferred from other observables (Salt Financial, 2021).

2 Data handling and preprocessing

2.1 Sampling

To construct the volatility forecasts for each stock respectively, the analysis first defines the appropriate sample-split scheme. To keep this report brief, it adopts a simple expanding window approach. At each forecast date, the model is estimated using only information that would have been available at that period of time. This "walk-forward" procedure prevents look ahead bias and limits overfitting by simulating how the model would be used in real time (Salt Financial, 2021). In this example it uses forecasted values in the forecasting structure, creating a model-based simulation of the one-month-ahead volatility path.

2.2 Data cleaning

The initial step in the data preparation process, involves a careful examination at the available high-frequency data. Although the dataset contains high-frequency data at the minute interval, they are not equally informative. Liu *et al.* (2015) documented that the simple five minute realised volatility estimate, provides a robust trade off between noise due to micro-structure distortions and information obtained (Andersen, 2000; Liu *et al.,* 2015). The realised volatility is also a more precise measure of latent volatility than alternatives like the squared return and therefore acts as a proxy for true volatility (Hansen and Lunde, 2011)

Therefore the original dataset is refined by only retaining the fifth minute observation within each intra-day interval, consistent with the practise of constructing the realised volatility measure. A detailed inspection of the resulting price series reveals no evidence of extreme outliers or obvious data recording errors. Moreover, the data exhibit a balanced structure across assets, with each stock containing an identical number of observations for prices, see Appendix A.

3 Estimation approach

One key insight Hansen and Lunde (2011) concluded was that high-frequency based realised volatility measures facilitated a better understanding of latent volatility and its dynamic properties, alongside the fact that more complex volatility models can benefit from high-frequency data.

3.1 Realised volatility

To estimate an assets latent volatility, the realised variance (RV) is calculated first. As latent volatility is not directly observable, the RV provides a proxy by aggregating squared intra-day log returns and capturing the cumulative variation of prices over a period. This makes it a valuable input for volatility forecasting models.

Suppose that the five minute price P_t is observed for $i = 1, 2, \ldots, n_t$ at trading day t:

$$r_{t,i} = \log\left(\frac{P_{t,i}}{P_{t,i-1}}\right) \tag{1}$$

Then the daily RV becomes the sum of the squared intra-day log returns:

$$RV_t = \sum_{i=1}^{n_t} r_{t,i}^2 \tag{2}$$

The realised volatility is then simply calculated as the square root of the RV.

3.2 HAR-RV

The primary forecasting model used in this report will be the heterogeneous autoregressive model for realised volatility (HAR-RV) (Corsi, 2009). This model offers a flexible approach to model realised volatility at the high frequency rate, is robust to short samples and acts as a baseline in the volatility forecasting literature (Salt Financial, 2021). The standard HAR-RV model is defined as:

$$RV_t = \beta_0 + \beta_1 RV_{t-1}^d + \beta_2 RV_{t-1}^w + \beta_3 RV_{t-1}^m + u_t \tag{3}$$

where β_j $(j = 0, 1, 2, 3)$ are unknown parameters that need to be estimated and RV_{t-1}^d, RV_{t-1}^w and RV_{t-1}^m denote the daily, weekly, and monthly lagged RV respectively. Specifically, RV_{t-1}^w and RV_{t-1}^m are denoted as follows:

$$RV_{t-1}^w = \frac{1}{4}\sum_{i=2}^{5} RV_{t-i} \tag{4}$$

$$RV_{t-1}^m = \frac{1}{17}\sum_{i=6}^{22} RV_{t-i} \tag{5}$$

This differs from the standard methodology as the HAR-RV specification in this report only spans unique information in each term, which has been shown to produce superior results (Salt Financial, 2021).

Volatility is then denoted as:

$$\sigma_t = \sqrt{RV_t} \tag{6}$$

See Appendix B for the time-series and descriptive statistics of the realised volatility.

3.3 Unit root

An unit root test is conducted to check stationarity. This is conducted to see if distribution of the past will continue in the future.

Table 1: Unit root test, realised volatility

Time horizon	p-value
Daily	0.00
Weekly	0.00
Monthly	0.02

The Augmented Dickey-Fuller (ADF) test suggests stationarity at the conventional 5% level for all time horizons. Suggesting that the time-series' fluctuate around a stable mean over time. This supports the use of linear HAR-RV framework.

4 Results

The results for the forecasted one-month-ahead volatility are documented in Figure 1.

Figure 1: One-month-ahead realised volatility forecasts

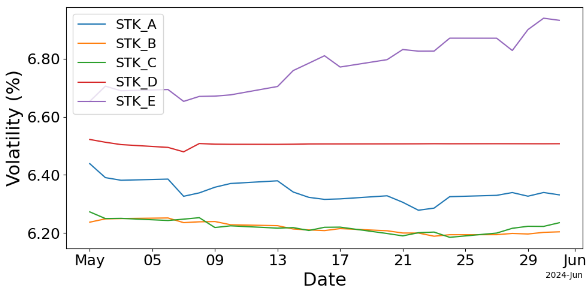


Figure 1 illustrates notable divergence in the one-month-ahead volatility forecasts across assets. For the majority of assets, the forecasted realised volatility evolves smoothly over time and remains relatively stable throughout the forecast horizon, indicating persistent volatility dynamics. In contrast STK_E shows a pronounced upward trend, suggesting elevated risk expectations over the forecast horizon. This pattern is corroborated by the descriptive statistics in Table 2, where STK_E displays both a higher mean level of forecasted volatility and a substantially larger standard deviation relative to the other assets.

Table 2: Descriptive statistics of one-month-ahead annualised realised volatility

Symbol	Count	Mean	Std.	Min	Median	Max
STK_A	23	6.34	0.04	6.28	6.33	6.44
STK_B	23	6.22	0.02	6.19	6.21	6.25
STK_C	23	6.22	0.02	6.18	6.22	6.27
STK_D	23	6.51	0.01	6.48	6.51	6.52
STK_E	23	6.78	0.09	6.65	6.78	6.94

Taken together, these results indicate that STK_E is characterised by greater variability, while the remaining assets exhibit more stable and predictable volatility dynamics over the one-month-ahead horizon.

4.1 Sanity checks

To assess the validity of the findings presented above, the one-month-ahead volatility forecasts are compared against the observed dynamics of the realised volatility during the preceding period. This comparison provides a diagnostic check on whether the model-implied forecasts are consistent with the historical volatility behaviour.

This comparison indicates that, for the majority of assets, the realised volatility remains relatively stable over the preceding period, exhibiting no pronounced trend or structural change prior to the forecast horizon. In contrast, STK_E displays a distinct and persistent upward trend in realised volatility in the months leading up to the one-month-ahead forecast period. This elevated and increasing realised volatility is subsequently reflected in the model's forecasts, suggesting that the HAR-RV framework effectively captures the volatility dynamics. See Appendix C.

4.2 Limitations and future research

The primary limitations of the forecasted volatility estimates lie with the underlying model specifications. In its standard form, the HAR-RV model relies exclusively on lagged realised variance measures to capture volatility dynamics across multiple time horizons. This implies that potentially informative explanatory variables are omitted. In particular, the model does not explicitly account for trading activity which could be closely related to volatility dynamics.

This can be simply incorporated by augmenting the standard HAR-RV model by adding a volume based explanatory variable in the regression.

Another limitation of the implemented HAR-RV model is the overnight jump. As the RV calculated in this report is strictly intra-day and therefore it does not take into account the contribution of the overnight jump, which can cause the model to systematically underestimate the volatility (Salt Financial, 2021).

This can be likewise addressed by extending the model to the HAR-J specification, which simply incorporates the previous day's overnight return as an added term in the regression (Salt Financial, 2021).

References

Andersen, T. G. (2000). Some reflections on analysis of high-frequency data. *Journal of Business & Economic Statistics*, *18*(2), 146–153.

Barchetto, A., Poirier, R., & Bae, J. (2021). *The layman's guide to volatility forecasting: Predicting the future, one day at a time* (Research Note). Salt Financial.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, *7*(2), 174–196.

Hansen, P. R., & Lunde, A. (2011). Forecasting volatility using high-frequency data. In *The oxford handbook of economic forecasting*. Oxford University Press.

Liu, L. Y., Patton, A. J., & Sheppard, K. (2015). Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes. *Journal of Econometrics*, *187*(1), 293–311.

Appendix

A Descriptive statistics

Figure 2: Time-series of prices

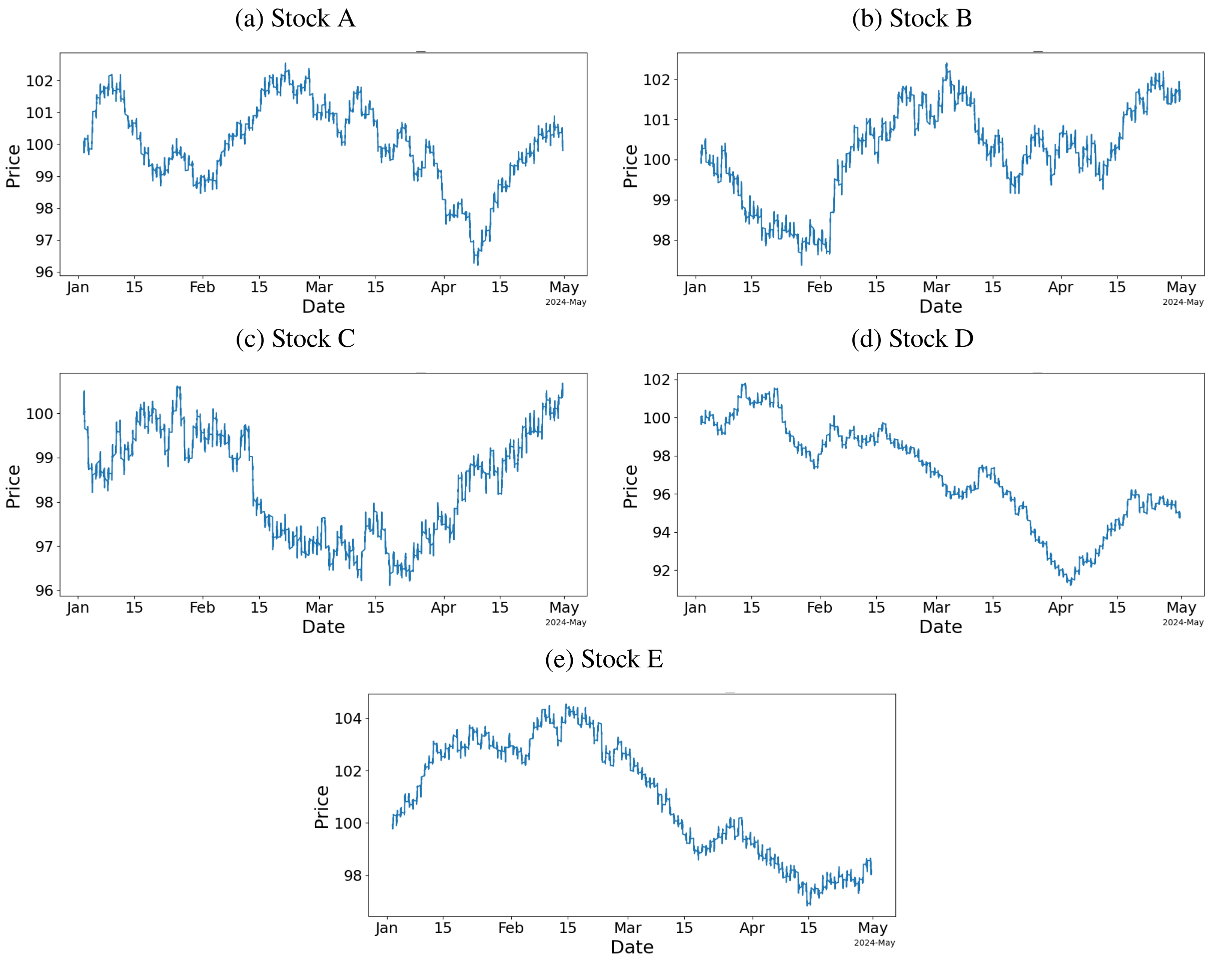


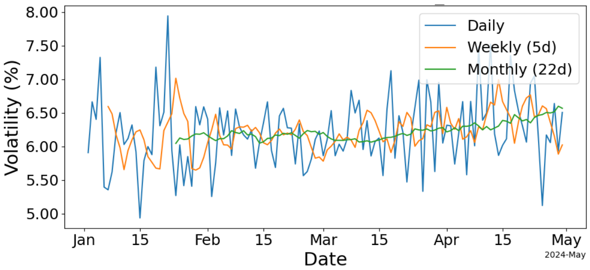
Table 3: Descriptive statistics of prices

Symbol	Count	Mean	Std. Dev.	Min	Median	Max
STK_A	9 384	100.01	1.30	96.21	100.06	102.54
STK_B	9 384	100.17	1.17	97.37	100.29	102.40
STK_C	9 384	98.41	1.20	96.11	98.65	100.68
STK_D	9 384	96.98	2.64	91.20	97.15	101.80
STK_E	9 384	100.96	2.22	96.83	101.08	104.54

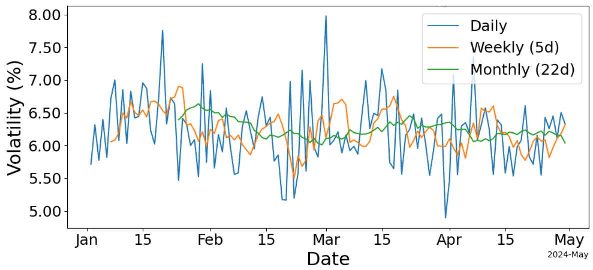
B Realised volatility time-series

Figure 3: Time-series of annualised realised volatility

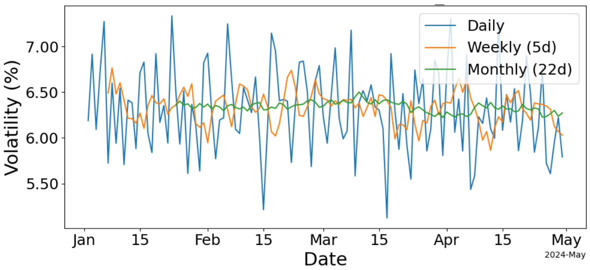
(a) Stock A



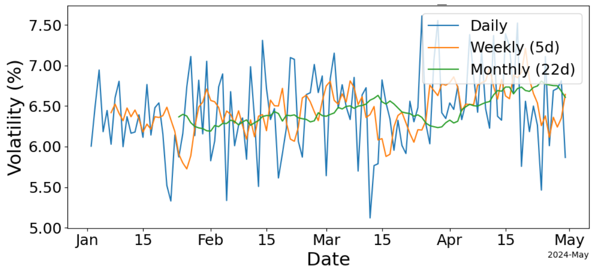
(b) Stock B



(c) Stock C



(d) Stock D



(e) Stock E

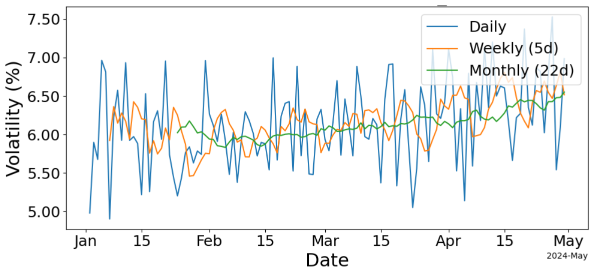


Table 4: Descriptive statistics of annualised realised volatility

Daily realised volatility						
Symbol	Count	Mean	Std.	Min	Median	Max
STK_A	120	6.21	0.52	4.94	6.17	7.94
STK_B	120	6.22	0.53	4.90	6.23	7.98
STK_C	120	6.31	0.46	5.12	6.30	7.33
STK_D	120	6.43	0.50	5.12	6.43	7.61
STK_E	120	6.15	0.56	4.90	6.16	7.53
Weekly realised volatility (5 trading days)						
STK_A	115	6.23	0.28	5.65	6.23	7.02
STK_B	115	6.24	0.28	5.50	6.20	6.90
STK_C	115	6.34	0.18	5.86	6.36	6.76
STK_D	115	6.45	0.27	5.72	6.44	7.21
STK_E	115	6.16	0.29	5.46	6.15	6.81
Monthly realised volatility (22 trading days)						
STK_A	98	6.22	0.12	6.05	6.19	6.60
STK_B	98	6.25	0.14	6.01	6.21	6.64
STK_C	98	6.34	0.06	6.22	6.34	6.50
STK_D	98	6.45	0.16	6.19	6.41	6.81
STK_E	98	6.13	0.17	5.83	6.10	6.55

C Sanity check

Figure 4: Time-series of annualised realised volatility + one-month-ahead